Venkata Sai

Software Engineer

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Professional Experience

AI Platform Engineer *Jul 2025 - Present*

Scale AI

- Built production **RAG** pipeline using **OpenAI** text embeddings and **FAISS** vector retrieval, enabling context-aware enterprise search across a 10M+ knowledge base with sub-500ms end-to-end latency.
- Designed **LangGraph** agentic workflows with human-in-the-loop oversight, routing low-confidence AI decisions to domain expert reviewers and reducing incorrect automation outcomes by 40% across production tasks.
- Engineered structured **prompt** templates with chain-of-thought reasoning and few-shot examples, improving **GPT-4o** output accuracy by 35% on classification and extraction tasks across five product surfaces.
- Implemented **Python** AI evaluation framework for hallucination detection, factuality scoring, and toxicity classification, enabling responsible AI quality controls and improvements across deployed product features.
- Fine-tuned instruction-tuned **LLM** with **LoRA** adapters and few-shot prompting, adapting base model behavior to domain-specific enterprise tasks and improving accuracy by 28% over zero-shot baseline.
- Architected **Python** LLM inference layer with token streaming, request batching, and latency-based routing, sustaining 5K+ daily users with 99.9% availability and sub-2-second p99 response times.
- Implemented dynamic **context management** for a multi-turn **LLM** assistant, chunking conversation history by token budget and pruning stale turns to maintain coherence across long sessions.
- Built **Python** A/B testing framework for **LLM** prompt variants, tracking quality and satisfaction metrics to enable data-driven iteration and safe rollback of underperforming model versions.

Software Engineer, AI *Jul 2022 - Jun 2025*

Palantir

- Integrated **OpenAI GPT-4** API into **React** and **Node.js** product surfaces, building streaming chat interfaces, inline AI suggestions, and structured output parsers for enterprise compliance workflows.
- Orchestrated **CrewAI** multi-agent pipeline assigning specialized agents to document parsing, validation, summarization, and escalation, automating 200+ daily enterprise compliance data ingestion, review, and triage.
- Built **Python** semantic search using **OpenAI** embeddings and **Pinecone** vector storage, enabling compliance officers to surface relevant policy documents three times faster than prior search.
- Trained **PyTorch** document classifier on 500K+ labeled compliance records, fine-tuning **BERT**-base transformer to achieve 91% F1 on enterprise regulatory category prediction with 40ms inference latency.
- Instrumented **Python** LLM monitoring pipeline capturing response quality, latency, and user feedback signals, powering an automated retraining feedback loop that improved output quality 22% quarterly.
- Built **AWS Lambda** pipelines ingesting 5M+ daily enterprise events into knowledge graphs, enabling AI compliance agents to query real-time relationship data with sub-second lookup latency.
- Applied **knowledge distillation** to compress a 7B-parameter **LLM** into a 500M student model, preserving 85% teacher accuracy and cutting inference cost by 60% in production.
- Built **responsible AI** bias evaluation pipeline using demographic parity and equalized odds metrics, detecting fairness violations before deploying **LLM** features to 10,000+ enterprise platform users.

Software Engineer *Jul 2020 - Aug 2022*

ThoughtWorks

- Developed **Python** REST APIs for a LabOps platform leveraging priority queues and graph algorithms for sample processing workflows, reducing average API response time by 35%.
- Authored optimized **PostgreSQL** queries and stored procedures for lab sample tracking, reducing report generation time by 30% through strategic index tuning and targeted query denormalization.
- Developed **Java Spring Boot** microservices alongside **Python** services in a multi-language stack, maintaining 75%+ test coverage with **JUnit** and **pytest** across all platform service boundaries.
- Containerized **Python** microservices using **Docker**, provisioned cloud environments with **Terraform**, and configured **GitHub Actions** CI/CD pipelines, reducing new-developer onboarding and environment provisioning time by 60%.

Skills

Languages: Python, Java, JavaScript, SQL
AI & ML: LLMs, RAG, embeddings, FAISS, Pinecone, PyTorch, BERT, LoRA, knowledge distillation, OpenAI API (GPT-4o), few-shot prompting, context management, agentic workflows, human-in-the-loop, responsible AI, bias mitigation, AI evaluation, A/B testing
Frameworks & Tools: LangGraph, CrewAI, React, Node.js, Spring Boot, pytest, JUnit
Cloud & Infra: AWS Lambda, Docker, Terraform, GitHub Actions
Databases: PostgreSQL

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India